11 R: The repository is the database: OntoRAG, ontology-governed retrieval over archaeological narrative sources

Marco Montanari (marco.montanari@gmail.com)

Most archaeological knowledge is narrative before it is data: excavation journals, context and tomb records, bibliography. Linked Open Data work usually begins after someone has already paid the cost of turning that prose into structure, which is why so many archives never enter the graph at all. Plain retrieval-augmented generation does not close the gap: it returns similar passages, and cannot answer "which tombs" or respect a controlled vocabulary.

OntoRAG is an Apache-2.0 suite that puts an ontology in front of retrieval and a person in front of the ontology. Its engine runs ingest → induce → align → extract → RDF; the Hub is a governance layer in which the proposed semantic model must be validated by a human before any extraction runs; an MCP server publishes any dataset to LLM clients; a desktop client runs the same pipeline on local compute. The dataset layout carries the argument: the git repository is the database - manifest, ontology, chunks and vectors as plain text, NDJSON and Turtle. Extraction runs in the dataset's own CI with the owner's own key, every stage commits its output, and publishing is a pull request. Versioning, diffs, rollback and audit come from git, not from a platform.

Semantically we align to the Open History Map ontology (w3id.org/ohm#, CC0): one canonical source emitted as OWL/Turtle, a JSON-LD context, SHACL shapes and JSON Schema, federated to Wikidata, PeriodO, GeoNames, GeoSPARQL and OWL-Time (not yet to CIDOC-CRM; we discuss why), with provenance - source, source type, reliability, author - required rather than optional.

We report work in progress on a real corpus: the excavation journals, feature records and bibliography of an early twentieth-century excavation, more than a thousand records in all, built into an OntoRAG dataset queryable by SPARQL and by a citing chatbot. We are explicit about the risk: an LLM-induced schema is a proposal, not a finding, and we discuss how often the human gate fires and what it catches.